CSP 203: Software Tools and Technologies (Question 1)

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1 Introduction

Live streaming is a powerful and direct way to get connected with thousands of concurrent viewers. No matter if you are using it for your product promotion, giving a motivating speech, or just want to stream the game.

1.1 Usage

It is a tool to circulate information in the form of videos. Users can enjoy the audio and video without waiting for their file to download.

1.2 Statistics

As live is on rise Halder et al. [2], the demand for high-quality and lower latency services is increasing. Thus, efficient, scalable, and economical solutions are required for live video streaming.

According to the report of Business Fortune Insights, the global video streaming market size is expected to grow from USD 342.44 billion in 2019 to USD 842.93 billion by 2027 at a compound annual growth rate of 12 Size [1].

2 Equation Section

The score of a node is calculated after taking uploading bandwidth, video streaming rate, latency, and stability of that node into consideration.

1. eqn. (1) has been proposed for the calculation of score.

$$Score(peer) = k_1*(\frac{upload_bandwidth}{streaming_rate})+k_2*(\frac{1}{latency})+k_3*(\frac{active_duration}{nums_of_failure}) \tag{1}$$

2. The general Tailor Series expansion is given in the form (2):

$$f(x) = f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\frac{f^{(3)}(a)}{3!}(x-a)^3+\cdots+\frac{f^{(n)}(a)}{n!}(x-a)^n+\cdots \tag{2}$$

3. The above expansion can be expressed as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

3 Table

Table 1: Performance Comparison of Algorithms

Algorithm	Time Complexity	Space Complexity
Linear Search	$O(n)$	$O(1)$
Binary Search	$O(\log n)$	$O(1)$
Merge Sort	$O(n \log n)$	$O(n)$
Quick Sort	$O(n \log n)$	$O(\log n)$

4 Algorithm

The following algorithm-(2) is used to search an element in a sorted array in $O(\log(n))$ time.

5 Image



Figure 1: Linux Desktop Image

The above fig.5 can be seen after we log in to the ubuntu operating system.

6 Bibliography

References

[1]

[2] Graphene Market Size. Video streaming market size. *Share & Trends Analysis Report by Streaming Type, by Solution, by Platform, by Service, by Revenue Model, by Deployment Type, by User and Segment Forecasts*, 2027, 2020.